

LAPORAN PRAKTIKUM
MATA KULIAH SISTEM TERDISTRIBUSI
PRAKTIKUM INSTALLASI HADOOP

Disusun sebagai bentuk pemenuhan tugas pada matakuliah system terdistribusi



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TAHUN 2026

ahap pertama setelah menjalankan sudo apt install openjdk-8-jdk:

```
ister [Running] - Oracle VirtualBox
Temporary failure resolving 'archive.ubuntu.com'
o fetch http://archive.ubuntu.com/ubuntu/pool/main/libi/libice/libice-dev_1.0.9-2ubuntu0.
64.deb Temporary failure resolving 'archive.ubuntu.com'
o fetch http://archive.ubuntu.com/ubuntu/pool/main/libp/libpthread-stubs/libpthread-stubs
_amd64.deb Temporary failure resolving 'archive.ubuntu.com'
o fetch http://archive.ubuntu.com/ubuntu/pool/main/libv/libvorbis/libvorbis0a_1.3.5-4.2_a
Temporary failure resolving 'archive.ubuntu.com'
o fetch http://archive.ubuntu.com/ubuntu/pool/main/libv/libvorbis/libvorbisenc2_1.3.5-4.2
Temporary failure resolving 'archive.ubuntu.com'
o fetch http://archive.ubuntu.com/ubuntu/pool/main/libs/libsndfile/libsndfile1_1.0.28-4ub
.2_amd64.deb Temporary failure resolving 'archive.ubuntu.com'
o fetch http://archive.ubuntu.com/ubuntu/pool/main/p/pulseaudio/libpulse0_11.1-1ubuntu7.1
Temporary failure resolving 'archive.ubuntu.com'
o fetch http://archive.ubuntu.com/ubuntu/pool/main/libs/libsm/libsm-dev_1.2.2-1_amd64.deb
Temporary failure resolving 'archive.ubuntu.com'
o fetch http://archive.ubuntu.com/ubuntu/pool/main/libx/libxau/libxau-dev_1.0.8-1ubuntu1_
Temporary failure resolving 'archive.ubuntu.com'
o fetch http://archive.ubuntu.com/ubuntu/pool/main/libx/libxdmcp/libxdmcp-dev_1.1.2-3_amd
Temporary failure resolving 'archive.ubuntu.com'
o fetch http://archive.ubuntu.com/ubuntu/pool/main/x/xtrans/xtrans-dev_1.3.5-1_all.deb T
Temporary failure resolving 'archive.ubuntu.com'
o fetch http://archive.ubuntu.com/ubuntu/pool/main/libx/libxcb/libxcb1-dev_1.13-2~ubuntu1
deb Temporary failure resolving 'archive.ubuntu.com'
o fetch http://archive.ubuntu.com/ubuntu/pool/main/libx/libx11/libx11-dev_1.6.4-3ubuntu0.
Temporary failure resolving 'archive.ubuntu.com'
o fetch http://archive.ubuntu.com/ubuntu/pool/main/libx/libx11/libx11-doc_1.6.4-3ubuntu0.
Temporary failure resolving 'archive.ubuntu.com'
o fetch http://archive.ubuntu.com/ubuntu/pool/main/libx/libxt/libxt-dev_1.1.5-1_amd64.deb
Temporary failure resolving 'archive.ubuntu.com'
o fetch http://archive.ubuntu.com/ubuntu/pool/universe/o/openjdk-8/openjdk-8-jre_8u372-ga
u1~18.04_amd64.deb Temporary failure resolving 'archive.ubuntu.com'
o fetch http://archive.ubuntu.com/ubuntu/pool/universe/o/openjdk-8/openjdk-8-jdk-headless
s1-0ubuntu1~18.04_amd64.deb Temporary failure resolving 'archive.ubuntu.com'
o fetch http://archive.ubuntu.com/ubuntu/pool/universe/o/openjdk-8/openjdk-8-jdk_8u372-ga
u1~18.04_amd64.deb Temporary failure resolving 'archive.ubuntu.com'
o fetch some archives, maybe run apt-get update or try with --fix-missing?
-master:~$ _
```

Berhasil download jdk:

```

update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/schemagen to provide /usr/bin/schemagen (schemagen) in auto mode
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/jps to provide /usr/bin/jps (jps) in auto mode
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/xjc to provide /usr/bin/xjc (xjc) in auto mode
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/jmap to provide /usr/bin/jmap (jmap) in auto mode
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/jstatd to provide /usr/bin/jstatd (jstatd) in auto mode
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/jdb to provide /usr/bin/jdb (jdb) in auto mode
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/serialver to provide /usr/bin/serialver (serialver) in auto mode
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/jfr to provide /usr/bin/jfr (jfr) in auto mode
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/wsgen to provide /usr/bin/wsgen (wsen) in auto mode
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/jcmd to provide /usr/bin/jcmd (jcmd) in auto mode
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/jarsigner to provide /usr/bin/jarsigner (jarsigner) in auto mode
Setting up openjdk-8-jre:amd64 (8u372-ga~us1-0ubuntu1~18.04) ...
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/jre/bin/policytool to provide /usr/bin/policytool (policytool) in auto mode
Setting up openjdk-8-jdk:amd64 (8u372-ga~us1-0ubuntu1~18.04) ...
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/appletviewer to provide /usr/bin/appletviewer (appletviewer) in auto mode
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/jconsole to provide /usr/bin/jconsole (jconsole) in auto mode
Processing triggers for libgdk-pixbuf2.0-0:amd64 (2.36.11-2) ...
Processing triggers for libc-bin (2.27-3ubuntu1.6) ...
andi@ubuntu-master:~$ java -version
openjdk version "1.8.0_362"
OpenJDK Runtime Environment (build 1.8.0_362-8u372-ga~us1-0ubuntu1~18.04-b09)
OpenJDK 64-Bit Server VM (build 25.362-b09, mixed mode)
andi@ubuntu-master:~$ _

```

Proses instalasi Hadoop:

```

andi@ubuntu-master:~$ wget https://archive.apache.org/dist/hadoop/common/hadoop-3.2.3/hadoop-3.2.3.tar.gz
--2026-05-04 08:50:55-- https://archive.apache.org/dist/hadoop/common/hadoop-3.2.3/hadoop-3.2.3.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 492241961 (469M) [application/x-gzip]
Saving to: 'hadoop-3.2.3.tar.gz'

hadoop-3.2.3.tar.gz      20%[=====>] 98.56M  203KB/s  eta 15m 30s

```

Berhasil install:

```

Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 492241961 (469M) [application/x-gzip]
Saving to: 'hadoop-3.2.3.tar.gz'

hadoop-3.2.3.tar.gz      100%[=====>] 469.44M   625KB/s   in 17m 1s

2026-05-04 09:07:57 (471 KB/s) - 'hadoop-3.2.3.tar.gz' saved [492241961/492241961]

andi@ubuntu-master:~$

```

Setelah menjalankan perintah tar -xvzf hadoop-3.2.3.tar.gz. Ini untuk ekstrak file :

```

Ubuntu Master [Running] - Oracle VirtualBox
hadoop-3.2.3/share/doc/hadoop/hadoop-distcp/images/maven-logo-2.gif
hadoop-3.2.3/share/doc/hadoop/hadoop-distcp/images/logo_apache.jpg
hadoop-3.2.3/share/doc/hadoop/hadoop-distcp/images/newwindow.png
hadoop-3.2.3/share/doc/hadoop/hadoop-distcp/images/icon_error_sml.gif
hadoop-3.2.3/share/doc/hadoop/hadoop-distcp/images/h5.jpg
hadoop-3.2.3/share/doc/hadoop/hadoop-distcp/images/icon_success_sml.gif
hadoop-3.2.3/share/doc/hadoop/hadoop-distcp/images/h3.jpg
hadoop-3.2.3/share/doc/hadoop/hadoop-distcp/DistCp.html
hadoop-3.2.3/share/doc/hadoop/hadoop-distcp/css/
hadoop-3.2.3/share/doc/hadoop/hadoop-distcp/css/maven-base.css
hadoop-3.2.3/share/doc/hadoop/hadoop-distcp/css/maven-theme.css
hadoop-3.2.3/share/doc/hadoop/hadoop-distcp/css/site.css
hadoop-3.2.3/share/doc/hadoop/hadoop-distcp/css/print.css
hadoop-3.2.3/share/doc/hadoop/hadoop-distcp/dependency-analysis.html
hadoop-3.2.3/lib/
hadoop-3.2.3/lib/native/
hadoop-3.2.3/lib/native/libhdfs.so.0.1.0
hadoop-3.2.3/lib/native/libhadooppipes.a
hadoop-3.2.3/lib/native/libhdfs.so.0.0.0
hadoop-3.2.3/lib/native/libhadooputils.a
hadoop-3.2.3/lib/native/libhadoop.so
hadoop-3.2.3/lib/native/libhdfs.so
hadoop-3.2.3/lib/native/libhadoop.so.1.0.0
hadoop-3.2.3/lib/native/libnativetask.a
hadoop-3.2.3/lib/native/examples/
hadoop-3.2.3/lib/native/examples/wordcount-part
hadoop-3.2.3/lib/native/examples/wordcount-nopipe
hadoop-3.2.3/lib/native/examples/pipes-sort
hadoop-3.2.3/lib/native/examples/wordcount-simple
hadoop-3.2.3/lib/native/libhdfs.a
hadoop-3.2.3/lib/native/libhdfs.so
hadoop-3.2.3/lib/native/libhadoop.a
hadoop-3.2.3/lib/native/libnativetask.so.1.0.0
hadoop-3.2.3/lib/native/libnativetask.so
hadoop-3.2.3/lib/native/libhdfs.a
hadoop-3.2.3/LICENSE.txt
andi@ubuntu-master:~$

```

Instalasi:

```

andi@LAPTOP-8B0CJ211: ~
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> wsl --install
Downloading: Ubuntu
Installing: Ubuntu
Distribution successfully installed. It can be launched via 'wsl.exe -d Ubuntu'
Launching Ubuntu...
Provisioning the new WSL instance Ubuntu
This might take a while...
Create a default Unix user account: andi
New password:
Retype new password:
passwd: password updated successfully
usermod: no changes
Help improve Ubuntu!

Help us improve Ubuntu features and compatibility by sharing system reports with Canonical.
Reports are sent anonymously and do not contain any personal data.
For legal details, please visit: https://ubuntu.com/legal/systems-information-notice

We will save your answer to Windows and will only ask you once.

Would you like to opt-in to platform metrics collection (Y/n)? To see an example of the data collected,
enter 'e'.
[Y/n/e]: y
andi@LAPTOP-8B0CJ211:/mnt/c/WINDOWS/system32$ ls
ls: cannot access '': No such file or directory
andi@LAPTOP-8B0CJ211:/mnt/c/WINDOWS/system32$ cd /mnt/c/Windows/System32
andi@LAPTOP-8B0CJ211:/mnt/c/Windows/System32$ find . -name *.dll | sort
./07409496-a423-4a3e-b620-2cfb01a9318d_HyperV-ComputeNetwork.dll
./0ae3b998-9a38-4b72-a4c4-06849441518d_Servicing-Stack.dll
./3bc29097-7317-41d3-93b9-38a48f99d48a_msrch.dll
./4545ffe2-0dc4-4df4-9d02-299ef204635e_hvsocket.dll
./5E374108-D6F1-471D-AE27-563CEAC0D6B2
./69fe178f-26e7-43a9-aa7d-2b616b672dde_eventlogservice.dll
./6bea57fb-8dfb-4177-9ae8-42e8b3529933_RuntimeDeviceInstall.dll

```

Praktikum:

Update ubuntu:

```

andi@LAPTOP-BB8CJ211:~$ sudo apt update && sudo apt upgrade -y
[sudo: authenticate] Password:
Get:1 http://archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [136 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security InRelease [136 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages
  [26.2 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main Translation-e
  n [14.1 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Compone
  nts [792 B]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main amd64 c-n-f M
  etadata [968 B]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Pac
  kages [34.0 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe Translati
  on-en [9728 B]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Co
  mponents [15.9 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-
  n-f Metadata [628 B]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64
  Components [212 B]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64
  c-n-f Metadata [120 B]
Get:14 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB
  ]
Get:15 http://archive.ubuntu.com/ubuntu resolute/main amd64 Packages [1480 k
  B]
16% [15 Packages 41.0 kB/1480 kB 3%]

```

Install java:

```
Processing triggers for dbus (1.10.2-2ubuntu4) ...
Processing triggers for install-info (7.2-5ubuntu2) ...
andi@LAPTOP-8B0CJ211:~$ sudo apt install openjdk-8-jdk -y
Installing:
  openjdk-8-jdk

Installing dependencies:
  alsa-topology-conf      libgtk2.0-common      libxdmcp-dev
  alsa-ucm-conf           libice-dev             libxft2
  ca-certificates-java   libice6               libxkbfile1
  fonts-dejavu-extra     libjpeg-turbo8        libxmu6
  gnome-accessibility-themes libjpeg8              libxpm4
  gnome-themes-extra     libmp3lame0           libxt-dev
  gnome-themes-extra-data libmpg123-0t64        libxt6t64
  gtk2-engines-pixbuf    libnspr4              libxv1
  ibus-gtk               libnss3               libxxf86dga1
  java-common            libogg0               linux-libc-dev
  libasound2-data        libopus0             luit
  libasound2t64          libpcscscit1         manpages-dev
  libasound2t64          libpulse0             openjdk-8-jdk-headless
  libatk-wrapper-java    libsm-dev             openjdk-8-jre
  libatk-wrapper-java-jni libsm6                openjdk-8-jre-headless
  libc-dev-bin           libsndfile1           rpcsvc-proto
  libc6-dev              libvorbis0a           uuid-dev
  libflac14              libvorbisenc2         x11-utils
  libgail-common         libx11-dev            x11proto-dev
  libgail18t64           libxau-dev            xorg-sgml-doctools
  libgif7                libxaw7               xtrans-dev
  libgtk2.0-0t64         libxcb-shape0
  libgtk2.0-bin          libxcb1-dev

Suggested packages:
  default-jre          pulseaudio             fonts-garum
```

Berhasil install java:

```
update alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/jconsole
provide /usr/bin/jconsole (jconsole) in auto mode
Processing triggers for libc-bin (2.43-2ubuntu2) ...
andi@LAPTOP-8B0CJ211:~$ java -version
openjdk version "1.8.0_482"
OpenJDK Runtime Environment (build 1.8.0_482-8u482-ga~us1-0ubuntu1-b08)
OpenJDK 64-Bit Server VM (build 25.482-b08, mixed mode)
andi@LAPTOP-8B0CJ211:~$
```

Download Hadoop:

```
andi@LAPTOP-8B0CJ211:~$ ls
andi@LAPTOP-8B0CJ211:~$ wget https://archive.apache.org/dist/hadoop/common/hadoop-3.2.3/hadoop-3.2.3.tar.gz
--2026-05-07 16:01:36-- https://archive.apache.org/dist/hadoop/common/hadoop-3.2.3/hadoop-3.2.3.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 492241961 (469M) [application/x-gzip]
Saving to: 'hadoop-3.2.3.tar.gz'

hadoop-3.2.3.tar.g  3%[          ] 17.71M  79.7KB/s  eta 68m 37s|
```

Progress: Instalasi Hadoop, masih stak di proses ini dikarenakan terkendala jaringan.

Lanjutan progress setelah berhasil install Hadoop:

Extract file dan rename:

```
hadoop-3.2.3/share/doc/hadoop/hadoop-distcp/dependenc
hadoop-3.2.3/lib/
hadoop-3.2.3/lib/native/
hadoop-3.2.3/lib/native/libhdfspp.so.0.1.0
hadoop-3.2.3/lib/native/libhadooppipes.a
hadoop-3.2.3/lib/native/libhdfs.so.0.0.0
hadoop-3.2.3/lib/native/libhadooputils.a
hadoop-3.2.3/lib/native/libhadoop.so
hadoop-3.2.3/lib/native/libhdfspp.so
hadoop-3.2.3/lib/native/libhadoop.so.1.0.0
hadoop-3.2.3/lib/native/libnativetask.a
hadoop-3.2.3/lib/native/examples/
hadoop-3.2.3/lib/native/examples/wordcount-part
hadoop-3.2.3/lib/native/examples/wordcount-nopipe
hadoop-3.2.3/lib/native/examples/pipes-sort
hadoop-3.2.3/lib/native/examples/wordcount-simple
hadoop-3.2.3/lib/native/libhdfs.a
hadoop-3.2.3/lib/native/libhdfs.so
hadoop-3.2.3/lib/native/libhadoop.a
hadoop-3.2.3/lib/native/libnativetask.so.1.0.0
hadoop-3.2.3/lib/native/libnativetask.so
hadoop-3.2.3/lib/native/libhdfspp.a
hadoop-3.2.3/LICENSE.txt
andi@LAPTOP-8B0CJ211:~$ mv hadoop-3.2.3 hadoop
```

Pindahkan ke folder sistem :

```
andi@LAPTOP-8B0CJ211:~$ sudo mv hadoop /usr/local/hadoop
[sudo: authenticate] Password:
andi@LAPTOP-8B0CJ211:~$ ls /usr/local/
bin  etc  games  hadoop  include  lib  libexec  man  sbin  share  src
andi@LAPTOP-8B0CJ211:~$ |
```

Edit file .bashrc:

```
bin  etc  games  hadoop  include  lib
andi@LAPTOP-8B0CJ211:~$ nano ~/.bashrc
```

Simpan ini dibagian paling bawah dan simpan:

```
elif [ -f /etc/bash_completion ]; then
. /etc/bash_completion
fi
# HADOOP VARIABLES START
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
export HADOOP_INSTALL=/usr/local/hadoop
export PATH=$PATH:$HADOOP_INSTALL/bin
export PATH=$PATH:$HADOOP_INSTALL/sbin
export HADOOP_MAPRED_HOME=$HADOOP_INSTALL
export HADOOP_COMMON_HOME=$HADOOP_INSTALL
export HADOOP_HDFS_HOME=$HADOOP_INSTALL
export YARN_HOME=$HADOOP_INSTALL
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_INSTALL/lib/native
export HADOOP_OPTS="-Djava.library.path=$HADOOP_INSTALL/lib/native"
export HADOOP_CONF_DIR=$HADOOP_INSTALL/etc/hadoop
# HADOOP VARIABLES END
```


Aktifkan konfigurasi dan tes:

```
andi@LAPTOP-8B0CJ211:~$ nano ~/.bashrc
andi@LAPTOP-8B0CJ211:~$ source ~/.bashrc
andi@LAPTOP-8B0CJ211:~$ echo $HADOOP_INSTALL
/usr/local/hadoop
andi@LAPTOP-8B0CJ211:~$ |
```

Masuk ke folder config dan Edit hadoop-env.sh:

```
andi@LAPTOP-8B0CJ211:/usr/local/hadoop/etc/hadoop$ nano hadoop-env.sh
andi@LAPTOP-8B0CJ211:/usr/local/hadoop/etc/hadoop$ |
```

```
# The java implementation to use. By default, this environment
# variable is REQUIRED on ALL platforms except OS X!
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64|
```

Edit core-site.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
  Licensed under the Apache License, Version 2.0 (the "License");
  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
  <name>fs.defaultFS</name>
  <value>hdfs://localhost:9000</value>
</property>
</configuration>
```

Edit mapred-site.xml:

```
te mapred-site.xml
nano mapred-site.xml
cp: cannot stat 'mapred-site.xml.template': No such file or directory
andi@LAPTOP-8B0CJ211:/usr/local/hadoop/etc/hadoop$ |
```



```
<configuration>
<property>
  <name>mapreduce.framework.name</name>
  <value>yarn</value>
</property>
</configuration>
```

Edit hdfs-site.xml:

```
<configuration>
<property>
  <name>dfs.replication</name>
  <value>1</value>
</property>
</configuration>
```

Edit yarn-site.xml

```
<configuration>|
<configuration>
  <property>
    <name>yarn.nodemanager.aux-services</name>
    <value>mapreduce_shuffle</value>
  </property>
</configuration>
</configuration>
```

Selesai config:

```
andi@LAPTOP-8B0CJ211: /usr, × + ▾
hadoop-3.2.3/lib/native/libnative-task.so
hadoop-3.2.3/lib/native/libhdfspp.a
hadoop-3.2.3/LICENSE.txt
andi@LAPTOP-8B0CJ211:~$ mv hadoop-3.2.3 hadoop
andi@LAPTOP-8B0CJ211:~$ sudo mv hadoop /usr/local/hadoop
[sudo: authenticate] Password:
andi@LAPTOP-8B0CJ211:~$ ls /usr/local/
bin etc games hadoop include lib libexec man sbin share src
andi@LAPTOP-8B0CJ211:~$ nano ~/.bashrc
andi@LAPTOP-8B0CJ211:~$ nano ~/.bashrc
andi@LAPTOP-8B0CJ211:~$ nano ~/.bashrc
andi@LAPTOP-8B0CJ211:~$ source ~/.bashrc
andi@LAPTOP-8B0CJ211:~$ echo $HADOOP_INSTALL
/usr/local/hadoop
andi@LAPTOP-8B0CJ211:~$ cd /usr/local/hadoop/etc/hadoop
andi@LAPTOP-8B0CJ211:/usr/local/hadoop/etc/hadoop$ nano hadoop-env.sh
andi@LAPTOP-8B0CJ211:/usr/local/hadoop/etc/hadoop$ nano core-site.xml
andi@LAPTOP-8B0CJ211:/usr/local/hadoop/etc/hadoop$ cp mapred-site.xml.templa
te mapred-site.xml
andi@LAPTOP-8B0CJ211:/usr/local/hadoop/etc/hadoop$ nano mapred-site.xml
cp: cannot stat 'mapred-site.xml.template': No such file or directory
andi@LAPTOP-8B0CJ211:/usr/local/hadoop/etc/hadoop$ nano hdfs-site.xml
andi@LAPTOP-8B0CJ211:/usr/local/hadoop/etc/hadoop$ nano yarn-site.xml
andi@LAPTOP-8B0CJ211:/usr/local/hadoop/etc/hadoop$ ls
capacity-scheduler.xml          kms-log4j.properties
configuration.xsl               kms-site.xml
container-executor.cfg          log4j.properties
core-site.xml                   mapred-env.cmd
hadoop-env.cmd                  mapred-env.sh
hadoop-env.sh                   mapred-queues.xml.template
hadoop-metrics2.properties      mapred-site.xml
hadoop-policy.xml               shellprofile.d
hadoop-user-functions.sh.examp
le                               ssl-client.xml.example
hdfs-site.xml                   ssl-server.xml.example
https-env.sh                    user_ec_policies.xml.template
https-log4j.properties          workers
https-signature.secret          yarn-env.cmd
https-site.xml                  yarn-env.sh
kms-acls.xml                    yarn-site.xml
kms-env.sh                      yarnservice-log4j.properties
andi@LAPTOP-8B0CJ211:/usr/local/hadoop/etc/hadoop$ |
```

Instal SSH:

```
andi@LAPTOP-8B0CJ211:/usr/local/hadoop/etc/hadoop$ sudo apt install ssh -y
[sudo: authenticate] Password:
Installing:
  ssh

Installing dependencies:
  libwrap0      openssh-server      ssh-import-id
  ncurses-term  openssh-sftp-server

Suggested packages:
  molly-guard  monkeysphere  ssh-askpass  ufw

Summary:
  Upgrading: 0, Installing: 6, Removing: 0, Not Upgrading: 0
  Download size: 1002 kB
  Space needed: 7737 kB / 1021 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libwrap0 amd64 7.
6.q-36build2 [48.5 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 ncurses-term all
6.6+20251231-1 [280 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 openssh-s
ftp-server amd64 1:10.2p1-2ubuntu3.2 [37.5 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 openssh-s
erver amd64 1:10.2p1-2ubuntu3.2 [622 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 ssh all 1
:10.2p1-2ubuntu3.2 [4662 B]
Get:6 http://archive.ubuntu.com/ubuntu resolute/main amd64 ssh-import-id all
5.11-0ubuntu4 [10.2 kB]
```

Jalankan service SSH, Generate SSH key, Aktifkan login tanpa password, dan tes ssh:

```
andi@LAPTOP-8B0CJ211:/usr/local/hadoop/etc/hadoop$ sudo service ssh start
andi@LAPTOP-8B0CJ211:/usr/local/hadoop/etc/hadoop$ ssh-keygen -t rsa -P "" -
 ~/.ssh/id_rsa
Generating public/private rsa key pair.
Created directory '/home/andi/.ssh'.
Your identification has been saved in /home/andi/.ssh/id_rsa
Your public key has been saved in /home/andi/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:x8IBzJ8Q2nr++KSb3MtEabT8C69XayKYaPbb4Ae+b9U andi@LAPTOP-8B0CJ211
The key's randomart image is:
---[RSA 3072]-----+
  oooo.
  o+  o
  . .oo.o
  . oo=
  . . * S.
  o.o .. E
  oo++...
  +o+%+oo.o
  o .@OX=oo
  ---[SHA256]-----+
andi@LAPTOP-8B0CJ211:/usr/local/hadoop/etc/hadoop$ cat ~/.ssh/id_rsa.pub >>
 ~/.ssh/authorized_keys
chmod 600 ~/.ssh/authorized_keys
andi@LAPTOP-8B0CJ211:/usr/local/hadoop/etc/hadoop$ ssh localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is: SHA256:BQJ6j6jKSc6YE7mf82kCJfR+a0dIVFr0JPI0J5idv
A
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_
4)
```

```
andi@LAPTOP-8B0CJ211:/usr/local/hadoop/etc/hadoop$ ssh localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is: SHA256:BQJ6j6jKSc6YE7mf82kCJfR+a0dIVFr0JPI0J5idv
A
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_
54)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Mon May 11 15:40:16 WIB 2026

System load:  0.0          Processes:    34
Usage of /:   0.5% of 1006.85GB  Users logged in: 0
Memory usage: 6%          IPv4 address for eth0: 172.17.231.201
Swap usage:   0%

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8
s
  just raised the bar for easy, resilient and secure K8s cluster deployment

  https://ubuntu.com/engage/secure-kubernetes-at-the-edge
andi@LAPTOP-8B0CJ211:~$ |
```

Jalankan hdfs namenode -format :

```
/usr/local/hadoop
andi@LAPTOP-8B0CJ211:~$ hdfs namenode -format
WARNING: /usr/local/hadoop/logs does not exist. Creating.
2026-05-11 15:46:36,539 INFO namenode.NameNode: STARTUP_MSG:
/*****
STARTUP_MSG: Starting NameNode
STARTUP_MSG:   host = LAPTOP-8B0CJ211/127.0.1.1
STARTUP_MSG:   args = [-format]
STARTUP_MSG:   version = 3.2.3
STARTUP_MSG:   classpath = /usr/local/hadoop/etc/hadoop:/usr/local/hadoop/sh
are/hadoop/common/lib/gson-2.2.4.jar:/usr/local/hadoop/share/hadoop/common/l
ib/zookeeper-3.4.14.jar:/usr/local/hadoop/share/hadoop/common/lib/jersey-cor
e-1.19.jar:/usr/local/hadoop/share/hadoop/common/lib/jackson-core-2.10.5.jar
:/usr/local/hadoop/share/hadoop/common/lib/kerb-server-1.0.1.jar:/usr/local/
hadoop/share/hadoop/common/lib/protobuf-java-2.5.0.jar:/usr/local/hadoop/sha
re/hadoop/common/lib/jetty-io-9.4.40.v20210413.jar:/usr/local/hadoop/share/h
adoop/common/lib/metrics-core-3.2.4.jar:/usr/local/hadoop/share/hadoop/commo
n/lib/jersey-server-1.19.jar:/usr/local/hadoop/share/hadoop/common/lib/commo
ns-math3-3.1.1.jar:/usr/local/hadoop/share/hadoop/common/lib/javax.activatio
n-api-1.2.0.jar:/usr/local/hadoop/share/hadoop/common/lib/avro-1.7.7.jar:/us
r/local/hadoop/share/hadoop/common/lib/jetty-server-9.4.40.v20210413.jar:/us
r/local/hadoop/share/hadoop/common/lib/jaxb-api-2.2.11.jar:/usr/local/hadoop
/share/hadoop/common/lib/guava-27.0-jre.jar:/usr/local/hadoop/share/hadoop/c
ommon/lib/jackson-annotations-2.10.5.jar:/usr/local/hadoop/share/hadoop/comm
on/lib/jersey-servlet-1.19.jar:/usr/local/hadoop/share/hadoop/common/lib/acc
essors-smart-2.4.7.jar:/usr/local/hadoop/share/hadoop/common/lib/snappy-java
-1.0.5.jar:/usr/local/hadoop/share/hadoop/common/lib/httpclient-4.5.13.jar:/
usr/local/hadoop/share/hadoop/common/lib/kerby-xdr-1.0.1.jar:/usr/local/hado
op/share/hadoop/common/lib/jetty-http-9.4.40.v20210413.jar:/usr/local/hadoo
p/share/hadoop/common/lib/failureaccess-1.0.jar:/usr/local/hadoop/share/hadoo
p/common/lib/jul-to-slf4j-1.7.25.jar:/usr/local/hadoop/share/hadoop/common/l
ib/kerb-client-1.0.1.jar:/usr/local/hadoop/share/hadoop/common/lib/jetty-uti
l-ajax-9.4.40.v20210413.jar:/usr/local/hadoop/share/hadoop/common/lib/commo
n-net-3.6.jar:/usr/local/hadoop/share/hadoop/common/lib/commons-text-1.4.jar
:/usr/local/hadoop/share/hadoop/common/lib/jaxb-impl-2.2.3-1.jar:/usr/local/
```

Start DFS:

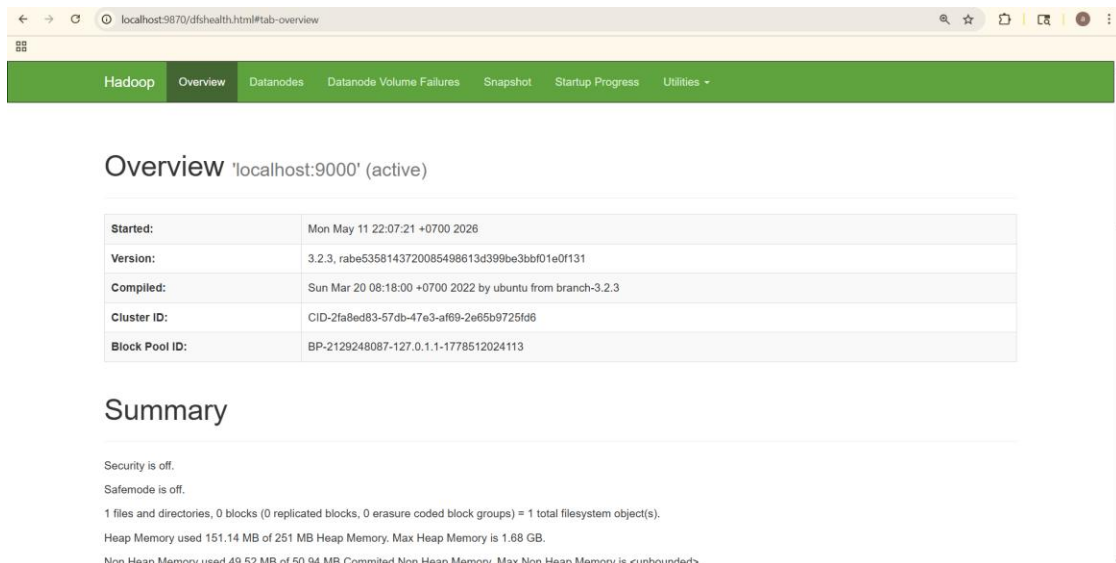
```
andi@LAPTOP-8B0CJ211:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [LAPTOP-8B0CJ211]
andi@LAPTOP-8B0CJ211:~$ jps
3249 NameNode
3397 DataNode
3736 Jps
3599 SecondaryNameNode
andi@LAPTOP-8B0CJ211:~$
```

Start yarn:

```
3599 SecondaryNameNode
andi@LAPTOP-8B0CJ211:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
andi@LAPTOP-8B0CJ211:~$ jps
3249 NameNode
3397 DataNode
4328 Jps
3849 ResourceManager
3995 NodeManager
3599 SecondaryNameNode
andi@LAPTOP-8B0CJ211:~$
```

Cek Hadoop di browser:

HDFS NameNode:



Overview 'localhost:9000' (active)

Started:	Mon May 11 22:07:21 +0700 2026
Version:	3.2.3, rabe5358143720085498613d399be3bbf01e0f131
Compiled:	Sun Mar 20 08:18:00 +0700 2022 by ubuntu from branch-3.2.3
Cluster ID:	CiD-2fa8ed83-57db-47e3-af69-2e65b9725fd6
Block Pool ID:	BP-2129248087-127.0.1.1-1778512024113

Summary

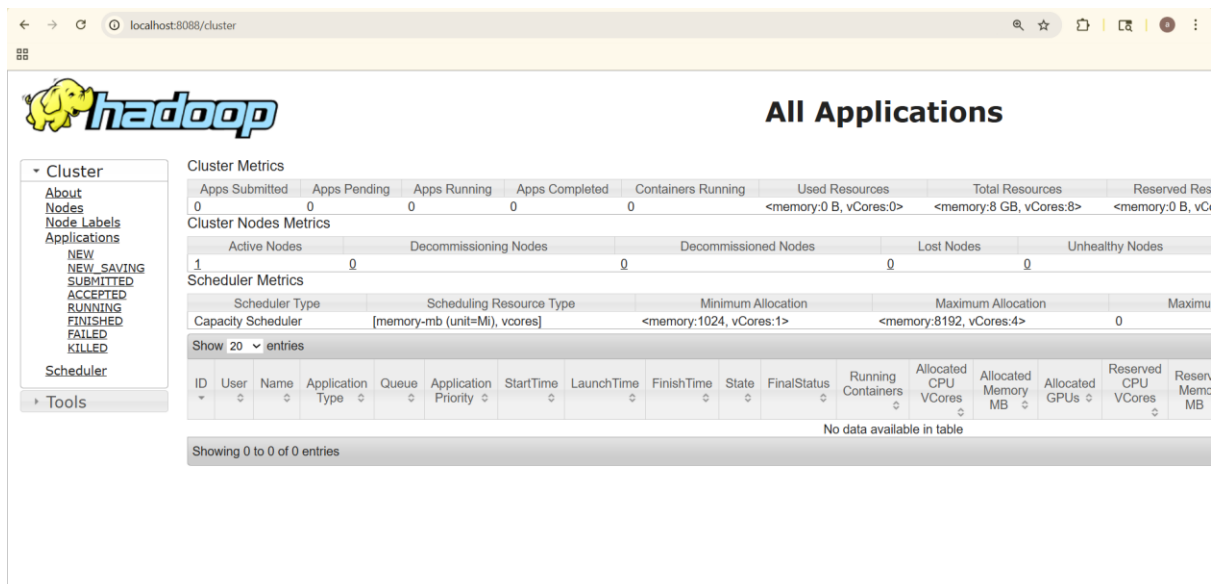
Security is off.
Safemode is off.

1 files and directories, 0 blocks (0 replicated blocks, 0 erasure coded block groups) = 1 total filesystem object(s).

Heap Memory used 151.14 MB of 251 MB Heap Memory. Max Heap Memory is 1.68 GB.

Non Heap Memory used 49.52 MB of 50.94 MB Committed Non Heap Memory. Max Non Heap Memory is <unbounded>.

YARN ResourceManager:



All Applications

Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	Used Resources	Total Resources	Reserved Resources
0	0	0	0	0	<memory:0 B, vCores:0>	<memory:8 GB, vCores:8>	<memory:0 B, vCores:0>

Cluster Nodes Metrics

Active Nodes	Decommissioning Nodes	Decommissioned Nodes	Lost Nodes	Unhealthy Nodes
1	0	0	0	0

Scheduler Metrics

Scheduler Type	Scheduling Resource Type	Minimum Allocation	Maximum Allocation	Maximum
Capacity Scheduler	[memory-mb (unit=Mi), vcores]	<memory:1024, vCores:1>	<memory:8192, vCores:4>	0

Show 20 entries

ID	User	Name	Application Type	Queue	Application Priority	StartTime	LaunchTime	FinishTime	State	FinalStatus	Running Containers	Allocated CPU VCores	Allocated Memory MB	Allocated GPUs	Reserved CPU VCores	Reserved Memory MB
No data available in table																

Showing 0 to 0 of 0 entries

Tes Hadoop sederhana:

```
3599 SecondaryNameNode
andi@LAPTOP-880CJ211:~$ hdfs dfs -mkdir /test
andi@LAPTOP-880CJ211:~$ hdfs dfs -ls /
Found 1 items
drwxr-xr-x - andi supergroup 0 2026-05-11 22:15 /test
andi@LAPTOP-880CJ211:~$
```

Hentikan Hadoop:

```
drwxr-xr-x - andi supergroup 0 2026-05-11 22:15 /test
andi@LAPTOP-8B8CJ211:~$ stop-dfs.sh
Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [LAPTOP-8B8CJ211]
andi@LAPTOP-8B8CJ211:~$ stop-yarn.sh
Stopping nodemanagers
Stopping resourcemanager
andi@LAPTOP-8B8CJ211:~$ jps
5227 Jps
andi@LAPTOP-8B8CJ211:~$
```

Analisis Praktikum:

Pada praktikum ini dilakukan instalasi dan konfigurasi Hadoop pada sistem Ubuntu menggunakan mode single-node cluster. Proses instalasi diawali dengan pengunduhan Hadoop, namun sempat mengalami kendala pada koneksi jaringan sehingga proses download dan pemindahan file Hadoop ke direktori /usr/local/hadoop berjalan sangat lama. Setelah itu dilakukan instalasi Java dan konfigurasi environment variable seperti JAVA_HOME, HADOOP_HOME, dan HADOOP_YARN_HOME. Selama proses konfigurasi muncul beberapa error seperti JAVA_HOME is not set, Invalid HADOOP_YARN_HOME, command not found, serta kesalahan konfigurasi fs.defaultFS yang menyebabkan NameNode gagal dijalankan. Kendala tersebut dapat diatasi dengan memperbaiki isi file konfigurasi Hadoop, menambahkan environment variable pada file .bashrc, melakukan source ~/.bashrc, serta memastikan struktur direktori Hadoop sudah benar.

Setelah seluruh konfigurasi diperbaiki, Hadoop berhasil dijalankan menggunakan perintah start-dfs.sh dan start-yarn.sh. Keberhasilan instalasi dibuktikan dengan munculnya proses NameNode, DataNode, SecondaryNameNode, ResourceManager, dan NodeManager pada hasil perintah jps. Selain itu dilakukan pengujian HDFS dengan membuat direktori menggunakan perintah hdfs dfs -mkdir /test dan pengecekan menggunakan hdfs dfs -ls /. Hasil pengujian menunjukkan bahwa folder berhasil dibuat sehingga menandakan sistem Hadoop dan HDFS telah berjalan dengan baik pada komputer lokal.

Kesimpulan

Berdasarkan praktikum yang telah dilakukan, dapat disimpulkan bahwa instalasi dan konfigurasi Hadoop single-node cluster berhasil dilakukan dengan baik meskipun terdapat beberapa kendala selama proses instalasi. Permasalahan yang muncul dapat diatasi dengan memperbaiki konfigurasi environment variable, file XML Hadoop, serta memastikan direktori dan layanan Hadoop berjalan dengan benar. Praktikum ini memberikan pemahaman mengenai

cara instalasi, konfigurasi, pengoperasian Hadoop, serta penanganan error yang sering terjadi pada proses setup Hadoop.